

Experiment 13

Problem Statement

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Algorithm

Step 1: Start

Step 2: Input integer n

Step 3: If $n \leq 0$

Print "Please enter a positive integer"

Stop

Step 4: Initialize:

$a \leftarrow 0$

$b \leftarrow 1$

~~seq~~ $\leftarrow$ empty list

Step 5: While $a \leq n$ do

Add a to ~~seq~~

$temp \leftarrow a + b$

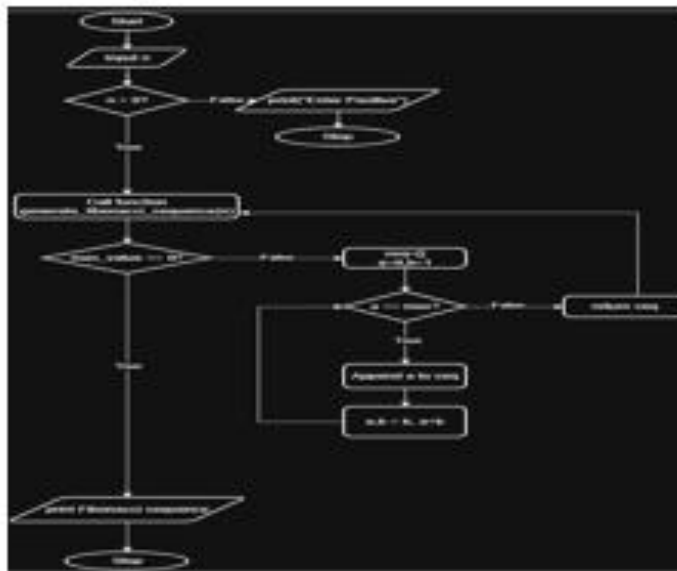
$a \leftarrow b$

$b \leftarrow temp$

Step 6: Print all elements of ~~seq~~

Step 7: Stop

Flowchart



Execution:

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12.1.1 Fibonacci Sequence using Modules

Write a Python program using module programming to generate the Fibonacci sequence up to a given maximum value

Module Requirements

- Create a module fibonacci_module.py with a function generate_fibonacci_sequence(max_value) that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create main.py that performs the following

- Import the fibonacci_module.
- Read an integer n from user input.
- Validate if n is positive.
- If valid, call the module function and display the Fibonacci sequence.
- If invalid, display an error message.

Input Format:

- Single integer n representing the maximum value for Fibonacci sequence.

Output Format:

- If n > 0, print the Fibonacci sequence as space-separated numbers.
- If n < 0, print, "Please enter a positive integer".

Note:

- Two files are provided: fibonacci_module.py and main.py. The main.py file handles input and output. You need to complete both files based on the question requirements.

main.py

```
1 import fibonacci_module
2 n = int(input())
3
4 if n > 0:
5     fib_series = fibonacci_module.generate_fibonacci_sequence(n)
6     print(" ".join(map(str, fib_series)))
7 else:
8     print("Please enter a positive integer")
```

12
1 1 2 3 5 8 13
== YOUR PROGRAM HAS COMPLETED ==